

The t - formulae 5C

1 a $2\sin\theta - \cos\theta = 2$

Using the substitution $t = \tan\frac{\theta}{2}$

$$2\left(\frac{2t}{1+t^2}\right) - \left(\frac{1-t^2}{1+t^2}\right) = 2$$

$$4t - 1 + t^2 = 2 + 2t^2$$

$$t^2 - 4t + 3 = 0$$

$$(t-3)(t-1) = 0$$

$$t = 3 \text{ or } t = 1$$

$$\tan\frac{\theta}{2} = 3$$

$$\frac{\theta}{2} = 1.1071\dots$$

$$\theta = 2.2142\dots$$

$$\text{or } \tan\frac{\theta}{2} = 1$$

$$\frac{\theta}{2} = \frac{\pi}{4}$$

$$\theta = \frac{\pi}{2} = 1.57$$

1 b $\sin\theta + 5\cos\theta = -1$

Using the substitution $t = \tan\frac{\theta}{2}$

$$\frac{2t}{1+t^2} - 5\left(\frac{1-t^2}{1+t^2}\right) = -1$$

$$2t + 5 - 5t^2 = -1 - t^2$$

$$4t^2 - 2t - 6 = 0$$

$$2t^2 - t - 3 = 0$$

$$(2t-3)(t+1) = 0$$

$$t = \frac{3}{2} \text{ or } t = -1$$

$$\tan\frac{\theta}{2} = \frac{3}{2}$$

$$\frac{\theta}{2} = 1.1071\dots$$

$$\theta = 2.2142\dots$$

$$\text{or } \tan\frac{\theta}{2} = -1$$

$$\frac{\theta}{2} = \frac{3\pi}{4}$$

$$\theta = \frac{3\pi}{2} = 4.71$$

1 c $\tan \theta - 5 \sec \theta = 7$

Using the substitution $t = \tan \frac{\theta}{2}$

$$\frac{2t}{1-t^2} - 5 \left(\frac{1+t^2}{1-t^2} \right) = 7$$

$$2t - 5 - 5t^2 = 7 - 7t^2$$

$$2t^2 + 2t - 12 = 0$$

$$t^2 + t - 6 = 0$$

$$(t+3)(t-2) = 0$$

$$t = -3 \text{ or } t = 2$$

$$\tan \frac{\theta}{2} = -3$$

$$\frac{\theta}{2} = \pi - 1.249 \dots$$

$$\theta = 3.79$$

$$\text{or } \tan \frac{\theta}{2} = 2$$

$$\frac{\theta}{2} = 1.107 \dots$$

$$\theta = 2.21$$

d $7 \cot \theta + 3 \operatorname{cosec} \theta = 9$

Using the substitution $t = \tan \frac{\theta}{2}$

$$7 \left(\frac{1-t^2}{2t} \right) + 3 \left(\frac{1+t^2}{2t} \right) = 9$$

$$7 - 7t^2 + 3 + 3t^2 = 18t$$

$$10 - 4t^2 = 18t$$

$$4t^2 + 18t - 10 = 0$$

$$2t^2 + 9t - 5 = 0$$

$$(2t-1)(t+5) = 0$$

$$t = \frac{1}{2} \text{ or } t = -5$$

$$\tan \frac{\theta}{2} = \frac{1}{2}$$

$$\frac{\theta}{2} = 0.436 \dots$$

$$\theta = 0.93$$

$$\text{or } \tan \frac{\theta}{2} = -5$$

$$\frac{\theta}{2} = \pi - 1.373 \dots$$

$$\theta = 3.54$$

e $2 \cot \theta - \operatorname{cosec} \theta = 0$

Using the substitution $t = \tan \frac{\theta}{2}$

$$2 \left(\frac{1-t^2}{2t} \right) + \left(\frac{1+t^2}{2t} \right) = 0$$

$$2 - 2t^2 - 1 - t^2 = 0$$

$$3t^2 - 1 = 0$$

$$t^2 = \frac{1}{3}$$

$$t = \frac{1}{\sqrt{3}} \text{ or } t = -\frac{1}{\sqrt{3}}$$

$$\tan \frac{\theta}{2} = \frac{1}{\sqrt{3}}$$

$$\frac{\theta}{2} = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{3} = 1.05$$

$$\text{or } \tan \frac{\theta}{2} = -\frac{1}{\sqrt{3}}$$

$$\frac{\theta}{2} = \pi - \frac{\pi}{6}$$

$$\theta = \frac{5\pi}{3} = 5.24$$

2 a $\sin 2\theta - 2 \cos 2\theta = 1 - \sqrt{3} \cos 2\theta$

Using the substitution $t = \tan \theta$

$$\frac{2t}{1+t^2} - 2 \left(\frac{1-t^2}{1+t^2} \right) = 1 - \sqrt{3} \left(\frac{1-t^2}{1+t^2} \right)$$

$$2t - 2 + 2t^2 = 1 + t^2 - \sqrt{3} + \sqrt{3}t^2$$

$$\sqrt{3}t^2 - t^2 - 2t + 3 - \sqrt{3} = 0$$

$$(\sqrt{3}-1)t^2 - 2t - (\sqrt{3}-3) = 0$$

$$2 \text{ b } t = \frac{2 \pm \sqrt{4 + 4(\sqrt{3} - 1)(\sqrt{3} - 3)}}{2(\sqrt{3} - 1)}$$

$$t = \frac{2 \pm \sqrt{28 - 16\sqrt{3}}}{2(\sqrt{3} - 1)}$$

You need to evaluate $\sqrt{28 - 16\sqrt{3}}$

Let $a + b\sqrt{3} = \sqrt{28 - 16\sqrt{3}}$, where a and b are integers.

$$\text{Then } (a + b\sqrt{3})^2 = 28 - 16\sqrt{3}$$

$$a^2 + 3b^2 + 2ab\sqrt{3} = 28 - 16\sqrt{3}$$

$$\text{So } a^2 + 3b^2 = 28 \text{ and } ab = -8$$

Solving simultaneously gives:

$$a^2 + 3\left(\frac{64}{a^2}\right) = 28$$

$$a^4 + 192 = 28a^2$$

$$a^4 - 28a^2 + 192 = 0$$

$$(a^2 - 16)(a^2 - 12) = 0$$

So $a^2 = 16$ (since a is an integer)

Therefore $a = \pm 4$

$$a = 4 \Rightarrow b = -2$$

$$a = -4 \Rightarrow b = 2$$

$$\text{Therefore } \sqrt{28 - 16\sqrt{3}} = 4 - 2\sqrt{3}$$

$$\text{or } \sqrt{28 - 16\sqrt{3}} = -4 + 2\sqrt{3}$$

$$\text{Now } t = \frac{2 \pm \sqrt{28 - 16\sqrt{3}}}{2(\sqrt{3} - 1)}$$

$$\text{So } t = \frac{2 + \sqrt{28 - 16\sqrt{3}}}{2(\sqrt{3} - 1)}$$

$$= \frac{2 + (4 - 2\sqrt{3})}{2(\sqrt{3} - 1)} = \frac{6 - 2\sqrt{3}}{2\sqrt{3} - 2}$$

$$= \frac{\sqrt{3}(2\sqrt{3} - 2)}{2\sqrt{3} - 2}$$

$$= \sqrt{3}$$

$$\text{or } t = \frac{2 + (-4 + 2\sqrt{3})}{2(\sqrt{3} - 1)} = \frac{-2 + 2\sqrt{3}}{2\sqrt{3} - 2} = 1$$

$$\text{Therefore } t = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3} \text{ or } \theta = \frac{4\pi}{3}$$

$$\text{and } t = 1 \Rightarrow \theta = \frac{\pi}{4} \text{ or } \theta = \frac{5\pi}{4}$$

$$3 \text{ a } 16\cot x - 9\tan x = 0$$

Using the substitution $t = \tan \frac{x}{2}$

$$16\left(\frac{1-t^2}{2t}\right) - 9\left(\frac{2t}{1-t^2}\right) = 0$$

$$16(1-t^2)^2 - 18t(2t) = 0$$

$$16(1-2t^2+t^4) - 36t^2 = 0$$

$$16 - 32t^2 + 16t^4 - 36t^2 = 0$$

$$16t^4 - 68t^2 + 16 = 0$$

$$4t^4 - 17t^2 + 4 = 0$$

$$\text{b Substituting } u = t^2$$

$$4u^2 - 17u + 4 = 0$$

$$(4u-1)(u-4) = 0$$

$$u = \frac{1}{4} \text{ or } u = 4$$

$$t^2 = \frac{1}{4} \text{ or } t^2 = 4$$

$$t = \frac{1}{2} \text{ or } t = -\frac{1}{2} \text{ or } t = 2 \text{ or } t = -2$$

$$\tan \frac{\theta}{2} = \frac{1}{2}$$

$$\frac{\theta}{2} = 0.463\dots$$

$$\theta = 0.93$$

$$\tan \frac{\theta}{2} = -\frac{1}{2}$$

$$\frac{\theta}{2} = \pi - 0.463\dots$$

$$\theta = 5.36$$

$$\tan \frac{\theta}{2} = 2$$

$$\frac{\theta}{2} = 1.107\dots$$

$$\theta = 2.21$$

$$\tan \frac{\theta}{2} = -2$$

$$\frac{\theta}{2} = \pi - 1.107\dots$$

$$\theta = 4.07$$

4 a $10\sin\theta\cos\theta - 3\cos\theta = -3$

Using the substitution $t = \tan\frac{\theta}{2}$

$$10\left(\frac{2t}{1+t^2}\right)\left(\frac{1-t^2}{1+t^2}\right) - 3\left(\frac{1-t^2}{1+t^2}\right) = -3$$

$$20t(1-t^2) - 3(1-t^4) = -3(1+t^2)^2$$

$$20t - 20t^3 - 3 + 3t^4 = -3(1+2t^2+t^4)$$

$$20t - 20t^3 - 3 + 3t^4 = -3 - 6t^2 - 3t^4$$

$$6t^4 - 20t^3 + 6t^2 + 20t = 0$$

$$3t^4 - 10t^3 + 3t^2 + 10t = 0$$

$$t(3t^3 - 10t^2 + 3t + 10) = 0$$

Let $f(t) = 3t^3 - 10t^2 + 3t + 10$

Then $f(2) = 3 \times 2^3 - 10 \times 2^2 + 6 + 10 = 0$

So $(t-2)$ is a factor of $f(t)$

$$3t^3 - 10t^2 + 3t + 10 = (t-2)(3t^2 + kt - 5)$$

Equating coefficients of t^2 :

$$-10 = k - 6$$

So $k = -4$

$$\therefore 3t^3 - 10t^2 + 3t + 10 = (t-2)(3t^2 - 4t - 5)$$

$$\therefore t(t-2)(3t^2 - 4t - 5) = 0$$

b So $t = 0$, $t = 2$, or $t = \frac{4 \pm \sqrt{76}}{6} = \frac{2 \pm \sqrt{19}}{3}$

$$\tan\frac{\theta}{2} = 0, \frac{\theta}{2} = 0$$

$$\theta = 0$$

$$\tan\frac{\theta}{2} = 2, \frac{\theta}{2} = 1.107\dots$$

$$\theta = 2.21$$

$$\tan\frac{\theta}{2} = \frac{2 + \sqrt{19}}{3}, \frac{\theta}{2} = 1.129\dots$$

$$\theta = 2.26$$

$$\tan\frac{\theta}{2} = \frac{2 - \sqrt{19}}{3}, \frac{\theta}{2} = \pi - 0.666\dots$$

$$\theta = 4.95$$

Substituting 2π into the original equation also gives -3 , so $\theta = 6.28$ (2 d.p.) is also

a solution.

5 a $3\sin 2\theta + \cos 2\theta + 3\tan 2\theta = 1$

Using the substitution $t = \tan\theta$

$$3\left(\frac{2t}{1+t^2}\right) + \frac{1-t^2}{1+t^2} + 3\left(\frac{2t}{1-t^2}\right) = 1$$

$$6t(1-t^2) + (1-t^2)^2 + 6t(1+t^2)$$

$$= (1+t^2)(1-t^2)$$

$$6t - 6t^3 + 1 - 2t^2 + t^4 + 6t + 6t^3 = 1 - t^4$$

$$2t^4 - 2t^2 + 12t = 0$$

$$t^4 - t^2 + 6t = 0$$

b $t^4 - t^2 + 6t = t(t^3 - t + 6)$

$(t+2)$ is a factor of $t^3 - t + 6$,

$$\text{So } t^3 - t + 6 = (t+2)(t^2 + kt + 3)$$

Equating coefficients of t^2 :

$$0 = 2 + k$$

So $k = -2$

Therefore $t(t+2)(t^2 - 2t + 3) = 0$

Note that $t^2 - 2t + 3 = 0$ has no solutions, since

$$'b^2 - 4ac' = (-2)^2 - 4 \times 1 \times 3 = -8 < 0$$

Therefore $t = 0$ or $t = -2$

$$\tan\theta = 0$$

$$\theta = 0, \theta = \pi \text{ or } \theta = 2\pi$$

$$\tan\theta = -2$$

$$\theta = \pi - 1.107\dots \text{ or } \theta = 2\pi - 1.107\dots$$

$$\theta = 2.03 \text{ or } \theta = 5.18$$

6 a $\tan\theta + \cos 2\theta = 1$

Using the substitution $t = \tan\theta$

$$t + \left(\frac{1-t^2}{1+t^2}\right) = 1$$

$$t(1+t^2) + 1 - t^2 = 1 + t^2$$

$$t + t^3 - t^2 = t^2$$

$$t^3 - 2t^2 + t = 0$$

6 b $t(t^2 - 2t + 1) = 0$

$$t(t-1)^2 = 0$$

$$\tan \theta = 0$$

$$\theta = 0 \text{ or } \theta = \pi \text{ or } \theta = 2\pi$$

$$\tan \theta = 1$$

$$\theta = \frac{\pi}{4} \text{ or } \theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

7 a $2\sin 2\theta - \cos 4\theta - 4\tan \theta = -1$

$$2\sin 2\theta - (\cos^2 2\theta - \sin^2 2\theta) - 4\tan \theta$$

$$= -1$$

$$2\sin 2\theta - \cos^2 2\theta + \sin^2 2\theta - 4\tan \theta$$

$$= -1$$

Using the substitution $t = \tan \theta$

$$2\left(\frac{2t}{1+t^2}\right) - \left(\frac{1-t^2}{1+t^2}\right)^2 + \left(\frac{2t}{1+t^2}\right)^2 - 4t = -1$$

$$\frac{4t(1+t^2) - (1-t^2)^2 + 4t^2 - 4t(1+t^2)^2}{(1+t^2)^2}$$

$$= -1$$

$$4t(1+t^2) - (1-t^2)^2 + 4t^2 - 4t(1+t^2)^2$$

$$= -(1+t^2)^2$$

$$4t + 4t^3 - (1 - 2t^2 + t^4) + 4t^2 - 4t(1 + 2t^2 + t^4)$$

$$= -(1 + 2t^2 + t^4)$$

$$4t + 4t^3 - 1 + 2t^2 - t^4 + 4t^2 - 4t - 8t^3 - 4t^5$$

$$= -1 - 2t^2 - t^4$$

$$-4t^5 - 4t^3 + 8t^2 = 0$$

$$t^5 + t^3 - 2t^2 = 0$$

b $t^5 + t^3 - 2t^2 = 0$

$$t^2(t^3 + t - 2) = 0$$

$$\text{Let } f(t) = t^3 + t - 2$$

$$f(1) = 1^3 + 1 - 2 = 0$$

So $t-1$ is a factor of $f(t)$

$$\text{So } f(t) = (t-1)(t^2 + kt + 2)$$

Equating coefficients of t^2

$$0 = -1 + k$$

$$k = 1$$

$$\text{Therefore } t^2(t-1)(t^2 + t + 2) = 0$$

Note that $t^2 + t + 2 = 0$

has no solutions, since

$$'b^2 - 4ac' = 1^2 - 4 \times 1 \times 2 = -7 < 0$$

So $t = 0$ or $t = 1$

$$\tan \theta = 0$$

$$\theta = 0, \theta = \pi \text{ or } \theta = 2\pi$$

$$\tan \theta = 1$$

$$\theta = \frac{\pi}{4} \text{ or } \theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

8 $5\cos\theta - 12\operatorname{cosec}\theta = 12$

Using the substitution $t = \tan \frac{\theta}{2}$

$$5\left(\frac{1-t^2}{1+t^2}\right) - 12\left(\frac{1+t^2}{2t}\right) = 12$$

$$5(2t)(1-t^2) - 12(1+t^2)^2 = 12(2t)(1+t^2)$$

$$10t(1-t^2) - 12(1+2t^2+t^4) = 24t + 24t^3$$

$$10t - 10t^3 - 12 - 24t^2 - 12t^4$$

$$= 24t + 24t^3$$

$$12t^4 + 34t^3 + 24t^2 + 14t + 12 = 0$$

$$6t^4 + 17t^3 + 12t^2 + 7t + 6 = 0$$

Let $f(t) = 6t^4 + 17t^3 + 12t^2 + 7t + 6$

$$f(-1) = 6 - 17 + 12 - 7 + 6 = 0$$

So $(t+1)$ is a factor of $f(t)$

Also $f(-2) = 96 - 136 + 48 - 14 + 6 = 0$

So $(t+2)$ is a factor of $f(t)$

Therefore $(t+1)(t+2) = t^2 + 3t + 2$

is a factor of $f(t)$

$$6t^4 + 17t^3 + 12t^2 + 7t + 6$$

$$= (t^2 + 3t + 2)(6t^2 + kt + 3)$$

Equate coefficients of t^2 :

$$12 = 3 + 3k + 12$$

So $k = -1$

So $(t+1)(t+2)(6t^2 - t + 3) = 0$

Note that $6t^2 - t + 3 = 0$

has no solutions, since

$$'b^2 - 4ac' = (-1)^2 - 4 \times 6 \times 3 = -71 < 0$$

So $t = -1$ or $t = -2$

$$\tan \frac{\theta}{2} = -1$$

$$\frac{\theta}{2} = \pi - \frac{\pi}{4}$$

$$\theta = \frac{3\pi}{2} = 4.71$$

$$\tan \frac{\theta}{2} = -2$$

$$\frac{\theta}{2} = \pi - 1.107\dots$$

$$\theta = 4.07$$

Challenge

$$5\sin 2\theta + 12\cos\theta = -12$$

$$10\sin\theta\cos\theta + 12\cos\theta = -12$$

Substitute $t = \tan \frac{\theta}{2}$

$$10\left(\frac{2t}{1+t^2}\right)\left(\frac{1-t^2}{1+t^2}\right) + 12\left(\frac{1-t^2}{1+t^2}\right) = -12$$

$$\frac{20t(1-t^2)}{(1+t^2)^2} + \frac{12(1-t^2)}{1+t^2} = -12$$

$$20t(1-t^2) + 12(1-t^2)(1+t^2) = -12(1+t^2)^2$$

$$20t - 20t^3 + 12(1-t^4) = -12(1+2t^2+t^4)$$

$$20t - 20t^3 + 12 - 12t^4 = -12 - 24t^2 - 12t^4$$

$$20t^3 - 20t - 24t^2 - 24 = 0$$

$$5t^3 - 6t^2 - 5t - 6 = 0$$

Let $f(t) = 5t^3 - 6t^2 - 5t - 6$

$$f(2) = 5 \times 2^3 - 6 \times 2^2 - 5 \times 2 - 6 = 0$$

So $(t-2)$ is a factor of $f(t)$

So $f(t) = (t-2)(5t^2 + kt + 3)$

Equating coefficients of t^2

$$-6 = -10 + k$$

$$k = 4$$

Therefore $(t-2)(5t^2 + 4t + 3)$

Note that $5t^2 + 4t + 3 = 0$

has no solutions, since

$$'b^2 - 4ac' = 4^2 - 4 \times 5 \times 3 = -34 < 0$$

So $t = 2$

$$\tan \frac{\theta}{2} = 2$$

$$\frac{\theta}{2} = 1.107\dots$$

$$\theta = 2.21$$

Check possible values of θ

for which $\tan\left(\frac{\theta}{2}\right)$ is undefined,

and which is in the given range $0 \leq \theta \leq 2\pi$

ie. $\theta = \pi$

This gives

$$5\sin 2\pi + 12\cos\pi = 5 \times 0 + 12 \times (-1) = -12$$

So $\theta = \pi$ is also a solution.